COURSE CODE: CSE402
COURSE TITLE: INTERNET OF THINGS LAB

PROJECT PROPOSAL

SUBMITTED TO:
Abrar Mahbub Tanim
Lecturer
Department of CSE
University of Information Technology & Sciences (UITS)

SUBMITTED BY:
Name: MD Fahad Rahman
ID: 0432320005101071
Section: 6B1
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1. Project Title

AirGuard IoT: An IoT-Based Smart Air Quality, Gas Leakage, and Environmental Monitoring System Using Arduino Uno and Cloud Computing

2. Domain

- **Primary Domain:** Environmental Monitoring & Public Health
- **Secondary Domain:** Smart Home & Industrial Safety (aligned with Domain-Specific IoT Applications)

3. Problem Statement

Poor indoor and ambient air quality, undetected hazardous gas leakages (e.g., LPG, carbon monoxide, methane), and oxygen depletion present acute safety hazards and long-term health risks in residential homes, institutional laboratories, and industrial workplaces. Traditional monitoring systems suffer from severe limitations:

- **Lack of Multi-Parameter Integration:** Conventional systems measure isolated parameters and fail to aggregate particulate dust, toxic gases, oxygen concentration, temperature, and humidity simultaneously.
- **No Remote Telemetry:** Existing smoke or gas alarms are standalone buzzer units lacking cloud connectivity or historical data logging for predictive safety assessments.
- **Delayed Emergency Response:** Without low-latency cloud notifications, stakeholders cannot respond to leaks or oxygen drops when off-premises.

4. Proposed Solution

AirGuard IoT delivers an edge-to-cloud environmental monitoring and early-warning architecture built upon the Arduino Uno R3 and ESP8266 Wi-Fi module.

The system continuously samples real-time physical telemetry using a specialized multi-sensor array: MQ-135 (air pollution/smoke), MQ-2 (combustible gas leakage), electrochemical O2 sensor (atmospheric oxygen percentage), Sharp GP2Y1010AU0F (airborne particulate matter), and DHT11/ DHT22 (temperature & humidity).

The Arduino Uno processes the conditioned sensor signals, refreshes an on-site 16x2 I2C LCD, and streams structured JSON payloads over serial communication to the ESP8266. The ESP8266 transmits this telemetry over 2.4 GHz Wi-Fi via MQTT/HTTP to an IoT cloud dashboard (ThingSpeak / Adafruit

IO). If any reading exceeds safety thresholds, the edge node instantly triggers an active buzzer and tri-color status LEDs while transmitting urgent alert flags to the cloud.

5. Objectives

- **Multi-Parameter Environmental Sensing:** Continuously measure Air Quality Index (AQI), hazardous gas PPM, O2 concentration, dust density (µg/m³), temperature, and humidity.
- **Early Leakage & Smoke Detection:** Rapidly detect combustible gases (LPG, methane) and smoke to prevent domestic fire and asphyxiation accidents.
- **Edge-to-Cloud Telemetry:** Establish reliable wireless data transmission between the Arduino Uno and cloud servers using the ESP8266 and MQTT protocol.
- **Live Visualization & Logging:** Display real-time numerical and graphical telemetry on a cloud dashboard for remote observation.
- **Dual-Layer Alarming:** Provide instantaneous on-site acoustic/visual alerts alongside remote cloud-based emergency notifications.

6. Hardware Requirements

- **Microcontroller Board:** Arduino Uno R3 (ATmega328P) for edge data acquisition and sensor control.
- **Wi-Fi Gateway Module:** ESP8266 (ESP-01 / NodeMCU) for 802.11 b/g/n Wi-Fi connectivity and MQTT messaging.
- **Air Quality Sensor (MQ-135):** Detects CO2, NH3, benzene, alcohol, and ambient smoke.
- **Combustible Gas Sensor (MQ-2):** Detects LPG, propane, methane, hydrogen, and combustible gas leaks.
- **Electrochemical O2 Sensor Module:** Measures ambient oxygen levels (safe threshold >= 19.5% O2).
- **Optical Dust Sensor:** Sharp GP2Y1010AU0F optical sensor for PM2.5 / PM10 particulate detection.
- **Climate Sensor (DHT11 / DHT22):** Digital sensor for relative humidity and ambient temperature.
- **Actuators & Display:** 16x2 Character LCD with I2C module, active 5V buzzer, and tri-color status LEDs (Green, Yellow, Red).
- **Power & Interface Accessories:** 5V/2A DC power supply, 3.3V logic level divider, breadboard, and jumper wires.

7. Software and Technologies

- **Firmware Development:** Arduino IDE (Embedded C/C++).
- **Firmware Libraries:** WiFiClient, PubSubClient (MQTT), DHT Sensor Library, LiquidCrystal_I2C, SoftwareSerial.
- **Communication Protocols:** MQTT (Message Queuing Telemetry Transport), HTTP REST API, UART Serial communication.
- **Cloud Platform:** ThingSpeak / Adafruit IO / Blynk IoT Cloud Dashboard.

8. Feasibility Analysis

- **Technical Feasibility:** The architecture uses established, standard embedded hardware components with rich library support. The Arduino Uno handles analog/digital sampling with high stability, while the ESP8266 offloads network tasks.
- **Economic Feasibility:** Built completely with low-cost commercial-off-the-shelf (COTS) modules, keeping prototype construction costs drastically lower than commercial industrial environmental analyzers.
- **Operational Feasibility:** Requires only standard 5V power and a standard 2.4 GHz Wi-Fi connection. Plug-and-play setup suitable for classrooms, homes, and computer laboratories with minimal

maintenance.

9. Project Scope and Limitations

- **Scope:** Indoor and localized laboratory/residential environmental monitoring, gas leakage safety alerting, cloud telemetry logging, and on-site visual reporting.
- **Hardware Limitations:** Arduino Uno has 2 KB SRAM and 32 KB flash memory; network intensive operations are delegated to the ESP8266 gateway.
- **Sensor Stabilization:** MQ-series metal-oxide sensors require an initial preheating burn-in period (~20-30 seconds) for stable baseline readings.
- **Network Dependency:** Cloud telemetry requires active Wi-Fi internet connectivity. However, local audio-visual alarm functions operate autonomously even when offline.

10. Expected Outcome

- **Functional Prototype:** A fully wired, tested hardware prototype integrating the Arduino Uno, ESP8266, and 5 environmental sensing modules.
- **Real-Time Cloud Dashboard:** An interactive web-based dashboard displaying live graphs and historical records for temperature, humidity, air quality, gas levels, oxygen, and dust.
- **Automated Alert Mechanism:** A dual-tier alarm system capable of sounding local buzzers and updating remote safety status within 2 seconds of detecting hazardous conditions.